

Preliminary Project Report 1:

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Topic: AI-Assisted Forensic Enhancement and Minutiae Localization in Partial Latent Fingerprints Using Pix2Pix Networks

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Github : <https://github.com/sdShaggy/GAFIS>

Outcomes expected : Presentable Literature + prototype

Abstract:

Latent fingerprints recovered from crime scenes are frequently incomplete, smudged, noisy, pressure-distorted, or partially destroyed due to environmental degradation and imperfect surface contact. Such degraded fingerprints significantly reduce the effectiveness of conventional Automated Fingerprint Identification Systems (AFIS), which are primarily optimized for high-quality rolled or sensor-acquired fingerprints.

This project proposes an AI-assisted forensic framework for the enhancement and analysis of partial latent fingerprints using a Conditional Generative Adversarial Network (Pix2Pix) architecture. The system is designed specifically for forensic assistance rather than biometric reconstruction, focusing on improving ridge continuity visibility, enhancing latent print usability, and assisting forensic examiners in minutiae localization and fingerprint comparison. The project aims to improve degraded latent fingerprint usability, reduce manual forensic workload, and demonstrate the effectiveness of AI-assisted enhancement techniques in forensic fingerprint analysis.

Literature Reviewed:

1) End-to-end latent fingerprint enhancement using multi-scale Generative Adversarial Network - Elsevier

By: Pramukha R.N.; Akhila P.; Shashidhar G. Koolagudi

This research proposed deep learning-based approach for enhancing latent fingerprints using a multi-scale Generative Adversarial Network (GAN). The method integrates Atrous Spatial Pyramid Pooling (ASPP), a multi-branch NN module designed to efficiently capture multi-scale contextual information for dense prediction tasks such as semantic image segmentation, to capture ridge details at multiple scales and improve the quality of noisy or partial fingerprints. Unlike traditional enhancement techniques, the proposed model focuses on reconstructing recoverable ridge patterns while minimizing the generation of false minutiae.

2) Latent fingerprint enhancement for accurate minutiae detection - Arxiv

By: Abdul Wahab, Tariq Mahmood Khan, Shahzaib Iqbal, Bandar AlShammari, Bandar Alhaqbani, Imran Razzak

This research focuses on improving latent fingerprint quality for reliable minutiae extraction using deep learning techniques. The authors proposed an enhancement framework designed to preserve ridge structures while reducing noise and distortions commonly present in latent fingerprints. The enhanced fingerprints significantly improved minutiae detection accuracy, which is essential for forensic identification systems. This novel approach too uses GANs for Latent Fingerprint Enhancement (LFE) through a structured approach to fingerprint generation. By directly optimising the minutiae information during the generation process, the model produces enhanced latent fingerprints that exhibit exceptional fidelity to ground-truth instances.

3) Latent Fingerprint Enhancement Using Generative Adversarial Networks - ACM/IEEE

By: Indu Joshi; Adithya Anand; Mayank Vatsa; Richa Singh; Sumantra Dutta Roy; Prem Kalra

The paper presented a deep learning-based framework for latent fingerprint enhancement and matching in forensic applications. The proposed approach utilized CNNs to improve ridge clarity, suppress background noise, and enhance minutiae extraction from poor-quality latent fingerprints. The study emphasized the importance of robust feature learning for improving fingerprint matching accuracy under challenging conditions such as partial prints, smudging, and low contrast. Experimental evaluations demonstrated that the deep learning model achieved better enhancement and recognition performance compared to traditional fingerprint preprocessing methods, highlighting its effectiveness for automated forensic identification systems.

4) Fingerprint Reconstruction and Identification using Convolutional Autoencoders - ACM

By: Parth Bhatnagar, Manjit Sodhi, Gururaj H Lokesh, Geethika Koravanghat, Dayananda Pruthviraja

This work propose and evaluate a framework employing Convolutional Neural Networks (CNNs) for robust fingerprint identification and Autoencoders for high-fidelity fingerprint reconstruction. The identification model demonstrated significant accuracy, effectively classifying genuine and various types of altered fingerprints, showcasing its resilience against deliberate manipulations. Concurrently, the autoencoder successfully restored degraded or partial fingerprint images, yielding high PSNR and SSIM scores, thereby enhancing the usability of compromised biometric data. This integrated approach not only improves the reliability of automated fingerprint identification systems (AFIS) against security threats but also provides a powerful tool for forensic analysis by recovering crucial pattern information.

5) Finger Minutiae Extraction Based on the use of YOLOv5 - ACM/Springer

By: Krzysztof Trusiak, Khalid Saeed

This work proposed a deep learning based approach for fingerprint minutiae detection using the YOLOv5 object detection framework. The method utilizes convolutional neural networks to automatically identify ridge endings and bifurcation points in fingerprint images with high precision and recall. Experimental results showed that the proposed model achieved effective minutiae localization and improved detection accuracy compared to conventional feature extraction techniques.

It is quite evident that most of the major researches implement a GAN in some way or another for latent fingerprint enhancement. GAN-based models have become widely adopted because of their ability to learn complex ridge structures and generate high-quality enhancements that preserve fine fingerprint details while suppressing background noise, compared to traditional noisy approaches like ridge frequency analysis. Many studies further integrate multi-scale feature extraction, encoder-decoder architectures, and object detection networks to improve ridge reconstruction and minutiae extraction accuracy.

Problem:

Modern forensic laboratories frequently encounter latent fingerprint evidence that is unsuitable for direct AFIS matching due to severe degradation and incomplete ridge structures. Major challenges include:

- Poor Ridge Visibility
- Contamination from Background
- Partial Fingerprint information
- Manual Forensic Dependence
- Reduced AFIS performance

Aim:

To develop an AI-assisted forensic fingerprint enhancement framework using Pix2Pix networks for improving ridge visibility, assisting minutiae localization, and enhancing latent fingerprint matching performance in forensic analysis workflows.

Proposed System Architecture:

1) Datasets:

SOCOFing	Training
FVC2004	Evaluation
NIST SD302 (subset)	Latent analysis

Artificial degradation techniques:

Gaussian blur, motion blur, smudging, contrast reduction, elastic distortion, partial masking, scratches

2) Preprocessing:

1. Grayscale normalization
2. Contrast enhancement
3. Ridge segmentation
4. Noise suppression
5. Gabor filtering
6. FFT-based enhancement

These operations will be done for enhancing ridge visibility, and preparing input for Pix2Pix Enhancement.

3) Ridge Orientation Estimation:

Orientation field estimation is used to preserve ridge flow consistency.

Techniques:

- gradient-based orientation estimation
- ridge coherence estimation
- directional smoothing

Output: Orientation map and Ridge flow visualization

4) Pix2Pix Based enhancement

Pix2Pix is a conditional Generative Adversarial Network (cGAN) designed for **paired image-to-image translation** tasks. It learns a direct mapping between an input image and a corresponding target image using a combination of an adversarial loss (to ensure realism) and a reconstruction loss (typically L1 loss to preserve structure). This makes it effective for tasks such as semantic segmentation map to photo translation, sketch-to-image synthesis, and image colorization.

Generator: It uses a UNet based CNN as a generator. The generator learns enhanced ridge continuity patterns from degraded latent fingerprints. Encoder compresses features, decoder reconstructs output. Skip connections preserve fine spatial details (important for edges/ridges).

Discriminator: PatchGAN, A CNN that classifies small patches of the image as real or fake. The discriminator differentiates between the real fingerprint ridge structures and AI-enhanced outputs.

The model does not reconstruct exact original fingerprints.

Instead, it predicts plausible ridge continuity patterns for forensic enhancement assistance.

5) Automated Minutiae Localization

YOLOv8 will be used for minutiae detection.

Minutiae extracted: Ridge endings and Bifurcations mainly, labelled properly

6) AFIS Matching Evaluation using SourceAFIS

The system compares:

- matching performance before enhancement
- matching performance after enhancement

This evaluates practical forensic usefulness.

Metrics: Following Metrics can be employed for evaluation of the prototype model.

1) SSIM (Structural Similarity Index Measure)

2) PSNR (Peak Signal to Noise Ratio)

3) Ridge continuity Index

4) Precision

5) Recall

6) F1 Score

7) Matching Score evaluation

8) FAR (False Acceptance Rate)

9) FRR (False Rejection Rate)

Tools and Tech Stack:

Programming - Python

Deep Learning Lib. - PyTorch

GAN - Pix2Pix Conditional GAN

Object Detection and Image Processing - YOLOv8 and OpenCV

Visualization - Matplotlib + Seaborn

AFIS - SourceAFIS

UI - Streamlit (Optional)

